MIL-V-60216 (MU)
3 MARCH 1965
SUPERSEDING
FA (FC)-PD-88
2 JULY 1961

MILITARY SPECIFICATION

VOLTAGE REGULATOR SUBASSEMBLY: 10525750

1. SCOPE

1.1 This specification covers one type of voltage regulator subassembly, a part of the Gun Direction Computer: M18. This voltage regulator subassembly contains the control elements of the plus 6-volt, the minus 3-volt and the minus 10-volt regulators.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926 — Fire Control Materiel, General Specification Governing the Manufacture and Inspection of.

MIL-P-14232 — Parts, Equipment and Tools for Ordnance Materiel; Packaging of.

MIL-I-45607 — Inspection Equipment, Supply and Maintenance of.

MIL-C-46357 — Computer, Gun Direction: M18; Manufacturing Processes and Inspection of.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-109 — Quality Assurance Terms and Definitions.

MIL-STD-202 — Test Methods for Electronic and Electrical Component Parts.

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

F10525750 — Voltage Regulator Subassembly.

PACKAGING DATA SHEET

M14002 — Data Sheet, Master Packaging.

FSC 1220

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:20Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, drawings, and packaging data sheets, required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. The voltage regulator shall be manufactured in accordance with Drawing F10525750 and drawings pertaining thereto.

3.2 General specifications. The voltage regulator shall meet the following requirements of MIL-F-13926, and MIL-C-46357.

3.2.1 *MIL-F-13926.*

(a) Order of precedence.

(b) Dimensions and tolerances.

(c) Inorganic protective surface finishes.

(d) Workmanship.

(e) Soldering electrical connections.

3.2.2 *MIL-C-46357.*

(a) Etched circuit assembly.

(b) Marking.

(c) Handling.

3.3 Environmental service conditions.

3.3.1 *Vibration.* The voltage regulator shall be subjected to a simple harmonic motion having an amplitude of 0.015 inch (0.030 inch total excursion). The frequency shall be varied uniformly from 10 to 55 cycles per second (cps) and return to 10 cycles. This cycle shall be traversed in 1 minute ± 15 seconds. The motion shall be applied to the voltage regulator for a period of 10 minutes in each direction of the 3 mutually perpendicular axes (total of 30 minutes), relative to its normal operational position. Subsequent to vibration, the voltage regulator shall not be damaged or adversely affected and shall meet the requirements of 3.4.

3.3.2 *Shock.* The voltage regulator shall withstand 3 shock impulses in the vertical plane relative to its normal mounting position. The acceleration of the shock impulses shall be 5 gravity units (G's). Pulse duration of each impulse shall be 11 ± 1 milliseconds. Subsequent to shock the item shall not have been damaged or adversely affected and shall meet the requirements of 3.4.

3.3.3 *Temperatures.*

3.3.3.1 *Operating.* The voltage regulator shall meet the requirements of 3.4.2 while exposed to temperatures of −40°F. and +135°F.

3.3.3.2 *Storage.* The voltage regulator shall meet the requirements of 3.4 at a temperature of between 60°F. and 90°F. after having been exposed to each temperature of −80°F. and +160°F.

3.3.4 *Humidity.*

3.3.4.1 *Operating.* The voltage regulator shall meet the requirements of 3.4.2 while exposed to an atmosphere with a relative humidity (RH) up to 97% at a temperature of 130°F.

3.3.4.2 *Nonoperating.* The voltage regulator shall meet the requirements of 3.4 after exposure to (and when thoroughly dried) an atmosphere with an RH of 100% including condensation, for a period of 4 hours.

3.3.5 *Fungus resistance.* The voltage regulator shall show no evidence of damage

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:20Z
Check the source to verify that this is the current version before use.

and shall meet the requirements of 3.4, after being innoculated with the composite spore suspension specified in MIL-F-13926 and incubated at an ambient temperature of between 77°F. and 84°F. with an RH of 97 ± 2%.

3.3.6 *Salt spray.* The voltage regulator shall show no evidence of physical damage and meet the requirements of 3.4 immediately after exposure to a fine spray of a 5% salt solution concentration. The solution shall have a specific gravity of from 1.0268 to 1.0413 and a pH between 6.5 and 7.2 at a temperature between 93° and 97°F.

3.4 Performance. Unless otherwise specified the voltage regulator shall meet the performance requirements of this specification at ambient temperatures between 60° and 90°F.; at a barometric pressure of 28 to 32 inches of mercury; and an RH up to 90%. Air shall be circulated across the components at the rate of 100 feet per minute (minimum), at the ambient temperature.

3.4.1 *Selection of resistors.* Resistors in table I shall be selected and assembled on the voltage regulator prior to meeting the subsequent requirements.

TABLE I. *Resistors used to set outputs*

Function	Resistors
+6 V Output votage	R66, R58.
+3V Output voltage	R21, R23.
−3V Output voltage	R25, R26.
−10V Output voltage	R69, R74.
+6V Trip current	R52, R43.
−10V Trip current	R8 , R9.

3.4.2 *Circuit output.* The circuits in this assembly are part of a regulated power supply and when connected to the appropriate external circuitry shall have Outputs Voltages as given in table II when the variable inputs and loads are those given in the same table.

3.4.3 *Resistors.* After final assembly into the voltage regulator, resistors which are not functionally part of the circuits referenced in 3.4.2 shall possess characteristics within tolerance limits specified on the appropriate drawings.

3.4.4 *Capacitors.* After final assembly into the voltage regulator, capacitors which are not functionally part of the circuits referenced in 3.4.2 shall possess characteristics within tolerance limits specified on the appropriate drawings.

3.4.5 *Diodes.* After final assembly into the voltage regulator, diode which are not functionally part of the circuits referenced in 3.4.2 shall possess characteristics within tolerance limits specified on the appropriate drawings.

3.4.6 *Transistors and silicon controlled rectifiers.* After final assembly into the voltage regulator, transistors and silicon controlled rectifiers which are not functionally part of the circuits referenced in 3.4.2 shall possess characteristics within tolerance limits specified on the appropriate drawings.

3.5 Electrical. Wiring and connections shall be in accordance with electrical schematic, Drawing F10525753.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:20Z
Check the source to verify that this is the current version before use.

TABLE II. *Inputs, loads, and outputs*

VARIABLE INPUTS Unless otherwise noted, all tolerances are ± 0.5 Volts.

Name	*Low line (volts)*	*Nominal line (volts)*	*High line (volts)*
+ 6V Main DC	+14.4	+17.1	+19.8
—10V Main DC	—15.0	—17.9	—20.8
+20V	+17.8	+21.8	+26.0
—32V	—25.0	—32.0	—37.5
—12V	—11.0 ± 0.2	—12.0 ± 0.2	—13.0 ± 0.2

LOADS Unless otherwise noted all tolerances are ± 5 percent.

Name	*Nominal load (amperes)*	*Minimum load (amperes)*	*Maximum load (amperes)*	*Trip load (amperes)*
+ 6V Load	0.35	0.25	0.45	
+ 3V Load	0.88	0.29	0.97	
— 3 V Load	0.49	0.15	0.63	
—10V Load	0.83	.36	1.30	1.5 ± 0.2
+ 6V Trip load				2.4 ± 0.2

Output voltages	*At terminal with respect to common*	*Nominal temperature D.C. volts*	*Ripple* P-P mv*	*Extreme temperature D.C. volts*	*Ripple* P-P mv*
+ 6V Output	59B	+6.00 ± 0.12	50	+6.00 ± 0.18	50
+ 3V Output	1B	+3.00 ± 0.12	50	+3.00 ± 0.18	50
— 3V Output	3B	—3.00 ± 0.12	50	—3.00 ± 0.18	50
—10V Output	37B	—10.00 ± 0.20	50	—10.00 ± 0.40	50

Tripped Output — All output voltage magnitudes less than 2.0 volt from common with all outputs having resistors across them equivalent (+ 10 percent) to the load current before trip.

* Ripple is herein defined as the peak-to-peak (p-p) value of any alternating component of the output voltage excluding noise of less than 10 microsecond duration.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *General provisions.* The component inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 *Submission of product.* Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 *Examinations and tests.* Except where otherwise specified herein, components and subassemblies shall be inspected in accordance with the "Inspection provisions for components and subassemblies" and the Sampling Plan Tables contained in

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:20Z
Check the source to verify that this is the current version before use.

MIL-F-13926. Examinations and tests related to section 3 herein shall be performed in accordance with tables III, IV and V and the test methods and procedures in 4.5. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and section 5 herein.

4.2.3 *Acceptance and rejection.* Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.2.4 *Functional tests.* The requirements and tests in table III shall be inspected on a 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (60° to 90°F.).

TABLE III. *(100 percent inspection)*

Characteristic	Requirement	Test procedures
101. Selection of resistors	3.4.1	4.5.3 through 4.5.3.5
102. Circuit output	3.4.2	4.5.4 through 4.5.4.6
103. Resistors	3.4.3	4.5.5
104. Capacitors	3.4.4	4.5.6
105. Diodes	3.4.5	4.5.7
106. Transistors and silicon controlled rectifiers	3.4.6	4.5.8
107. Electrical	3.4.7	4.5.9

4.3 Standard sampling.

4.3.1 *General.* One unit shall be selected at random by a Government representative as a standard sample from each 100 produced. The sample shall meet the requirements and tests in table IV.

TABLE IV.

Characteristic	Requirement	Test procedures
301. Fabrication	3.1	Applicable drawings — visual.
302. General specifications	3.2	MIL-F-13926 and MIL-C-46357 — visual.

4.3.2 *Environmental.* Three units shall be selected at random by a Government representative as a standard sample from each 50 produced or from each month's production, whichever occurs first. The samples shall have met the requirements and tests in table III and shall then meet the requirements and tests in table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:20Z
Check the source to verify that this is the current version before use.

TABLE V.

Characteristic	Requirement	Test procedures
303. Vibration	3.3.1	4.5.1.1
304. Shock	3.3.2	4.5.1.2
305. Humidity-Operating	3.3.4.1, 3.4.2	4.5.1.3.2, 4.5.4
306. Humidity-Nonoperating	3.3.4.2	4.5.1.3.1
307. Salt spray	3.3.6	4.5.1.4
308. Operating temperature	3.3.3.1, 3.4.2	4.5.1.5, 4.5.4
309. Storage temperature	3.3.3.2	4.5.1.5

4.3.3 *Failure of sample.* Should any one item of standard sampling fail to meet the specified test requirements, acceptance of the represented production lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.2.3).

4.4 **Inspection equipment.** Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will no unduly interfere with the contractor's delivery schedule.

4.4.1 *Contractor design.* The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolernace limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.4.1.1 *Test equipment.* In conjunction with 4.4.1, the following test equipment shall be utilized in the performance of the applicable tests as specified in 4.5:

Item number	Description
1. Voltage measuring device	0–50 vdc, input impedance 10 megohms or greater, accuracy of ± .5% or better.
2. Resistance measuring device	0–200 ohms, accuracy ± .5% of scale reading.
3. Current measuring device	0–10 amperes, accuracy ± .1% full scale reading.
4. Oscilloscope	Rise time of scope plus measuring probe, 50 nano seconds or less, sweep rate 50 nano seconds per centimeter, input impedance 8 megohms and 25 pico farads or greater.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:20Z
Check the source to verify that this is the current version before use.

In addition to the measuring devices specified above, the voltage regulator subassembly shall utilize the power supplies, signal sources and potentiometers outlined in tables I, II and VI through VIII and figures 1 through 6.

TABLE VI. *Power supplies*

*Three Phase Full Wave Bridge Types**

Identifying name	*Variable range of output (volts)*	*Nominal setting (volts)*	*Current rating (amperes)*
+ 6V Main DC	+ 12 to 22	+ 17.1	3.0
— 10V Main DC	— 12 to 24	— 17.9	2
+ 20V	+ 16 to 28	+ 21.8	2.0
— 32V	— 23 to 40	— 32.0	0.15

Regulated Types

Identifying name	*Variable range or tolerance*	*Maximum RMS ripple millivolts*	*Current amperes*
+ 15V	± 2 percent	2	0.025
+ 6V	± 2 percent	2	0.1
— 12V	— 11.0 to — 13.0 volts	2	0.1

*See figure 1 for characteristics of this type power supply.

TABLE VII. *Potentiometer connections.*

Potentiometer	*Quantity set*	*Type**	*+ Terminal*	*Terminal*	*Wiper terminal*	*Initial wiper setting from + terminal (ohms)*
1	— 10V Output	II	37A	34A	32A	35
2	+ 6V Output	II	10B	6A	13A	35
3	— 3V Output	II	12A	14B	4B	35
4	+ 3V Output	I	10A	11B	7B	25
5	— 10V Trip current	III	49B	38B	41B	105
6	+ 6V Trip current	III	25B	11A	33A	105

*Completion of fabrication of the voltage regulator requires the use of the following three types of potentiometers:

Type	*Total resistance ohms (± 1 percent)*	*Required range of wiper (ohms)*	*Current rating (amperes)*
I	50	5 to 45	.04
II	70	5 to 65	.04
III	110	5 to 105	.04

All types shall be capable of wiper settings in increments of 5 ohms ± 5 percent over the required wiper range. The potentiometer shall be used in such a way that the wiper is never open circuited. (The total resistance shall be within the 1 percent tolerance specified at all times when measurements are being made, but may have a ± 6 ohm tolerance at other times.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:20Z
Check the source to verify that this is the current version before use.

TABLE VIII. *Resistors*

Resistor numbers	Value ohms
10526702	5
10526703	6
10526704	6.7
10526705	8.4
10526706	10
10526707	11.7
10526708	13.4
10526709	15
10526710	16.7
10526711	18.4
10526712	20
10526713	21.7
10526714	23.4
10526715	25
10526716	26.7
10526717	28.4
10526718	30
10526719	35
10526720	40
10526721	45
10526722	50
10526723	55
10526724	60
10526725	65
10526726	70
10526727	75
10526728	80
10526729	85
10526730	90
10526731	95
10526732	100
10526733	105

4.5 Test methods and prcoedures.

4.5.1 *Environmental.*

4.5.1.1 *Vibration.* The vibration test shall be made on a device with means provided for measuring and controlling the direction of vibration and adjusting the amplitudes and frequencies to keep them within the specified limits of 3.3.1. The measuring instruments of the testing device shall be capable of measuring the vibration amplitude within ± 5 percent, and capable of measuring the frequency within ± 2 percent. The test shall be performed in accordance with the requirements specified in 3.3.1 with the unit rigidly mounted in its normal operating position. Subsequent to the vibration test, the unit shall show no damage or defect resulting from the test when examined by visual and tactile means. The unit shall then be inspected as specified in 4.5.1.2 prior to determining compliance with requirements specified under 3.4.

4.5.1.2 *Shock.* This test shall be conducted in accordance with Procedure II of the shock test defined in MIL-F-13926 except that the voltage regulator shall be shocked in the vertical plane only. The impact (shocks) and "G" value shall be in accordance with 3.3.2. A receiving device, representing the connector receptacle that receives the voltage regulator in the computer shall be secured to the horizontal surface of the shock tester. Each end of the receiving device shall be clamped such that the female receiving slot will hold the unit in a vertical plane. The unit will then be inserted into the receiving device and shocked. Subsequent to imposing the number of impacts required, a visual examination shall be made for damages or defects to determine compliance with 3.3.2 and then the unit will be inspected as specified in 4.5.1.3 prior to determining compliance with 3.4 as required by 3.3.2.

4.5.1.3 *Humidity.*

4.5.1.3.1 *Nonoperating.* This test shall be conducted in accordance with the humidity test defined in Procedure I of MIL-F-13926 except that the 4½ hour period shall be reduced to 2 hours, the 6 hour period reduced to 2 hours, and the test period shall not exceed 5.5 hours per cycle. The unit shall then be thoroughly dried and then subjected to the test in 4.5.1.3.2 prior to meeting the specified requirements of 3.4.

4.5.1.3.2 *Operating.* This test shall be conducted in accordance with the humidity test defined in Procedure III of MIL-F-13926 ex-

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:20Z
Check the source to verify that this is the current version before use.

cept that the humidity shall be maintained at 97 ± 5 percent for the 24 hour period, and the test shall cease at the end of that period (Method 106 of MIL-STD-202 shall be excluded). Throughout the test the unit shall be operated in accordance with 4.5.4 of this specification, and shall meet the specified requirements of 3.3.4.1.

4.5.1.4 *Salt spray.* The unit shall be subjected to the salt spray test defined in MIL-STD-202, Condition B. Immediately after exposure, the unit shall be inspected for any physical damage, then tested in accordance with 4.5.4 to 4.5.9 inclusive for compliance with 3.3.6.

4.5.1.5 *Temperature (operating and storage).* After being subjected to all environmental tests specified above, the unit shall be subjected to Test Procedure I specified in MIL-F-13926, except the temperatures specified in 3.3.3 shall apply. While at the operating temperatures of 3.3.3.1, the unit shall be tested in accordance with 4.5.4 to determine compliance with the requirements of 3.4.2 as specified in 3.3.3.1. The temperature shall then be returned to standard temperature (60° to 90°F.) and the unit shall be tested in accordance with 4.5.4 to 4.5.9 inclusive to determine compliance with 3.3.3.2.

4.5.1.6 *Fungus.* One of the first 5 units produced shall be submitted for testing in accordance with the fungus test defined in MIL-F-13926. After approval, subsequent acceptance for fungus requirements may be by certification.

4.5.2 *Performance.* To determine compliance with 3.4 the test methods and procedures in 4.5.3 through 4.5.3.5 shall be performed as specified.

4.5.3 *Selection of resistors.* The resistors listed in table I shall be selected as indicated herein so that the voltage regulator shall meet the tests of 4.5.4.1 to 4.5.4.5 inclusive.

4.5.3.1 *External circuitry.* Connect the external circuitry to the voltage regulator as shown in figures 2, 3, 4, 5, and 6. The external power supply characteristics shall be those of table VI and figure 1. Additional conditions for testing shall be as follows:

(a) Potentiometers – Connect the potentiometers described and as indicated in table VII.

(b) Loads – Adjust all loads listed in table II such that for nominal output voltages the load current conditions of all regulators shall not exceed those listed in the Maximum Current column or be less than those listed in the Minimum Current column.

(c) Power supply turn-on sequence – Energize the power supplies. Unless otherwise noted, the following sequence shall always be followed when energizing the power supplies: (1) Energize the –12v and –32v power supplies; (2) Energize the + 20v, + 6v, and + 15v power supplies; (3) Energize the + 6v Main dc and –10v Main dc power supplies.

4.5.3.2 *Output voltage resistors.* Adjust the potentiometers 1, 2, 3, and 4 listed in table VII such that the output voltages are as close to those listed in Output Voltage column of table V as the potentiometer resolution will allow. The output voltages shall be read to an accuracy of ± 0.1 percent. The Variable Inputs and Loads shall be those listed in the Nominal Line and Maximum Load columns of table II.

4.5.3.3 *Minus 10v trip current resistors.* At the conditions defined in the Nominal Line and Nominal Load columns of table II, adjust potentiometer 5 listed in table VII such that when the –10v load is increased

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:20Z
Check the source to verify that this is the current version before use.

the voltage regulator trips when the load is as close to 1.5 amperes as the resolution of the potentiometer allows. The Tripped Output is defined under Output Voltages in table V. The load current shall be measured to an accuracy of ± 25 milliamperes. At no time shall the –10v load current exceed 1.7 amperes. After the voltage regulator is tripped, it may be reset by de-energiizng all power supplies and, with proper loads, re-energizing them in proper sequence.

4.5.3.4 *Plus 6v trip current resistors.* At Nominal Load columns of table II, adjust conditions defined in the Nominal Line and potentiometer 6 listed in table VII such that when the current from A to A′ (figure 2) is as close to 2.4 amperes as the resolution of the potentiometer setting allows. The resistance from A to A′ shall not exceed 0.6 ohm. The current shall be read to an accuracy of ± 25 milliamperes. At no time shall the current exceed 2.6 amperes.

4.5.3.5 *Installation of resistors.* Complete the fabrication of the voltage rgeulator by installing the resistors listed in table I, using resistors listed in table VIII. The resistor values shall be those determined by the potentiometer settings for the respective resistors.

4.5.4 *Circuit outputs.* To determine compliance with 3.4.2, tests shall be performed as specified in 4.5.4.1 through 4.5.4.6.

4.5.4.1 *External circuitry.* Connect the external circuitry to the voltage regulator as shown in figures 2, 3, 4, 5, and 6. The power supply characteristics shall be those listed in table VI and figure 1. Additional conditions for testing shall be as follows:

(a) Loads – Adjust all loads specified in table II such that for nominal output voltages the load current conditions shall not exceed those listed in the Maximum Current column or be less than those listed in the Minimum Current column.

(b) Power supply turn-on sequence – Energize the power supplies. Unless otherwise noted, the following sequence shall always be followed when energizing the power supplies: (1) Energize the –12v and –32v power supplies; (2) Energize the + 20v, + 6v, and + 15v power supplies; (3) Energize the + 6v Main dc and –10v Main dc power supplies.

4.5.4.2 *Output voltage.* The output voltages of the voltage regulator shall be the Output Voltage Nominal Temperature column of table II when the conditions of the Variable Inputs and Loads of table II are combined in the following column pairs:

(a) Low Line – Maximum Load.

(b) Low Line – Minimum Load.

(c) High Line – Minimum Load.

(d) High Line – Maximum Load.

4.5.4.3 *Minus 10v overload trip current.* At conditions defined in the Nominal Line and Maximum Load columns, increase the –10v load. The voltage regulator shall trip at the –10v Trip Load specified in table II. The –10v load shall never exceed 1.75 amperes. The voltage regulator Tripped Output is defined under Voltage Outputs in table II. The voltage from terminal 44B to common shall be 0 to + 2 volts. After the voltage regulator is tripped, it may be reset by de-energizing all power supplies and, with proper loads, re-energizing them in proper sequence.

4.5.4.4 *Plus 6v overload trip current.* At conditions defined in the Nominal Line and

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:20Z
Check the source to verify that this is the current version before use.

Maximum Load columns of table II, increase the + 6v load current. The voltage regulator shall trip when the current from A to A′ (fig. 2) is the + 6v Trip Load specified in table II. The resistance from A to A′ shall not exceed 0.6 ohm. The current shall not exceed 2.65 amperes. The voltage from 22B to common shall be 0 to + 2.0 volts.

4.5.4.5 *Marginal test.* Perform the following at the conditions listed in the Nominal Line and Nominal Load columns of table II:

(a) Disconnect terminals 35A and 24B from common. The −10v output shall decrease in magnitude 5 ± 1.0 percent. The + 6v output shall increase in magnitude 5 ± 1.0 percent. Reconnect terminals 35A and 24B to common.

(b) Connect terminals 30A and 20B to common. The −10v output shall increase in magnitude 5 ± 1.0 percent. The + 6v output shall decrease in magnitude 5 ± 1.0 percent. Disconnect terminals 30A and 20B from common.

4.5.4.6 *Remote sensing.* At conditions defined in the Nominal Line and Nominal Load columns of table II, disconnect terminals 17B, 19B, 5A, 1B, 3B, 33B, and 37B from the common. The −10 volt, −3 volt and + 3 volt output voltages shall change no more than 0.5 volt from the output voltage measured before these terminals were disconnected. The + 6 volt output voltage shall change no more than 0.6 volt from the output voltage measured before these terminals were disconnected. Reconnect the terminals.

4.5.5 *Resistors.* The resistance between terminals 57A and 65A shall be 80 ohms ± 5% to determine compliance with 3.4.3.

4.5.6 *Capacitors.* Capacitors specified in 3.4.4 shall be tested as outlined on the appropriate drawings.

4.5.7 *Diodes.* Diodes specified in 3.4.5 shall be tested as outlined on the appropriate drawings.

4.5.8 *Transistors and silicon controlled rectifiers.* Transistors used in external circuitry as called for in figures 2, 3, 4, and 5 shall not exceed a case temperature of 110°F. and shall be capable of passing the electrical requirements of their respective drawings to insure cmopliance with 3.4.6.

4.5.9 *Electrical.* The voltage regulator shall be visually inspected to insure that all components are physically present with all wiring and connection in accordance with Drawing F10525753 as required by 3.5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, pcakaging and packing. Preservation, packaging and packing shall be in accordance with Master Packaging Data Sheet M14002.

6. NOTES

6.1 Intended use. The voltage regulator subassembly contains the control elements of a plus 6-volt series type voltage regulator, a plus 3-volt shunt type voltage regulator, a minus 3-volt shunt type voltage regulator and a minus 10-volt series type voltage regulator. Provisions are made for fail-safe remote sensing at the load and for increasing or decreasing the output voltages by five percent for marginal testing procedures. Temperature compensated overcurrent protection is provided for the plus 6-volt and minus 10-volt regulators, when overcurrent conditions are detected a signal is sent to an external master trip curcuit that shuts down selected regulators. Efficiency is increased by directing the plug 6-volt load current through the minus 3-volt shunt regulator and its load.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:20Z
Check the source to verify that this is the current version before use.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Selection of applicable level of preservation, packaging and packing.

(c) Applicable stock number.

(d) Applicable packaging data sheet (see 5.1).

(e) A block of serial numbers to identify each voltage regulator.

Custodian:

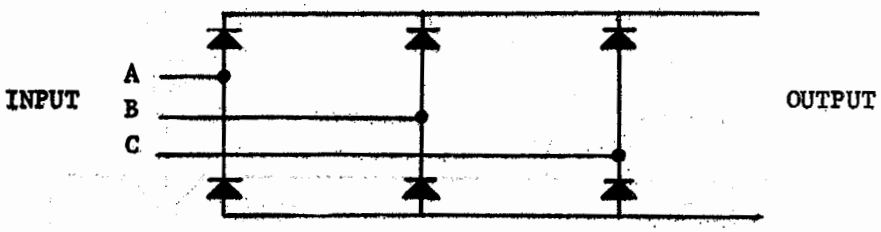
Army—MU

Preparing activity:

Army—MU

Project No. 1220-A198Y

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:20Z
Check the source to verify that this is the current version before use.



INPUT: Frequency: 400 ± 20 cycles per second

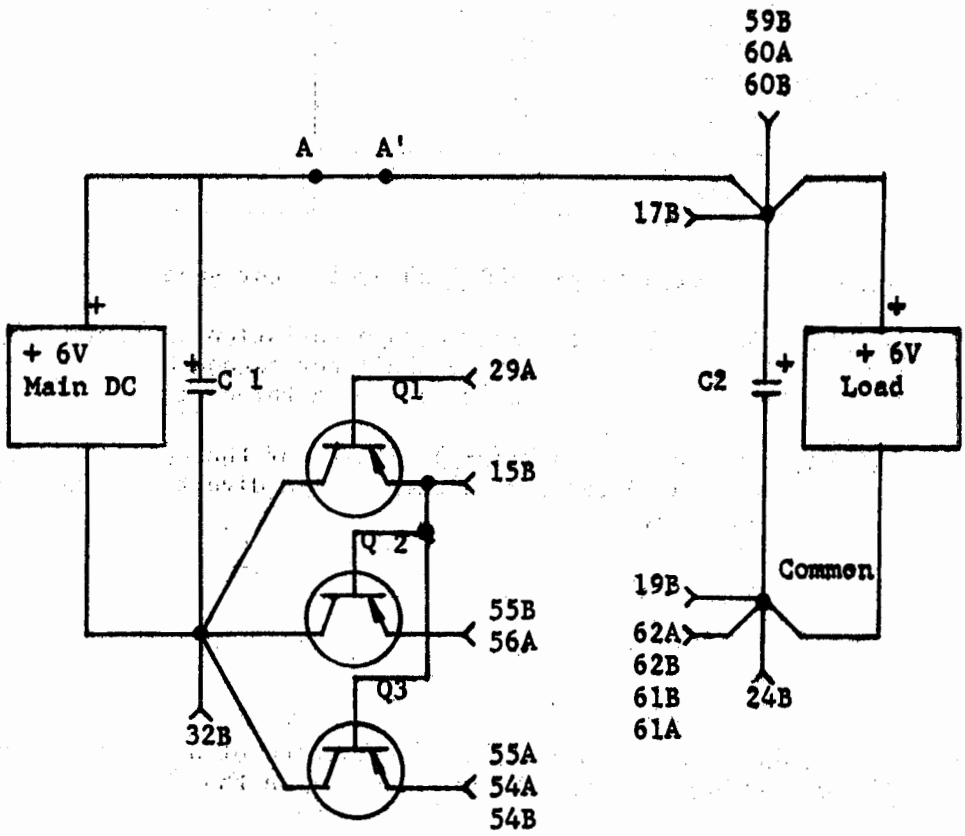
Line Balance: The voltage variation between any two lines shall not exceed 8 percent of the average of the three line to line voltages.

OUTPUT: The peak-to-peak (p-p) ripple* of the output shall not exceed 20 percent of the average direct current output voltage.

* Ripple is herein defined as peak-to-peak (p-p) value of any alternating component of the Output Voltage excluding noise of less than 10 microsecond duration.

FIGURE 1. *Three phase full wave bridge.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:20Z
Check the source to verify that this is the current version before use.

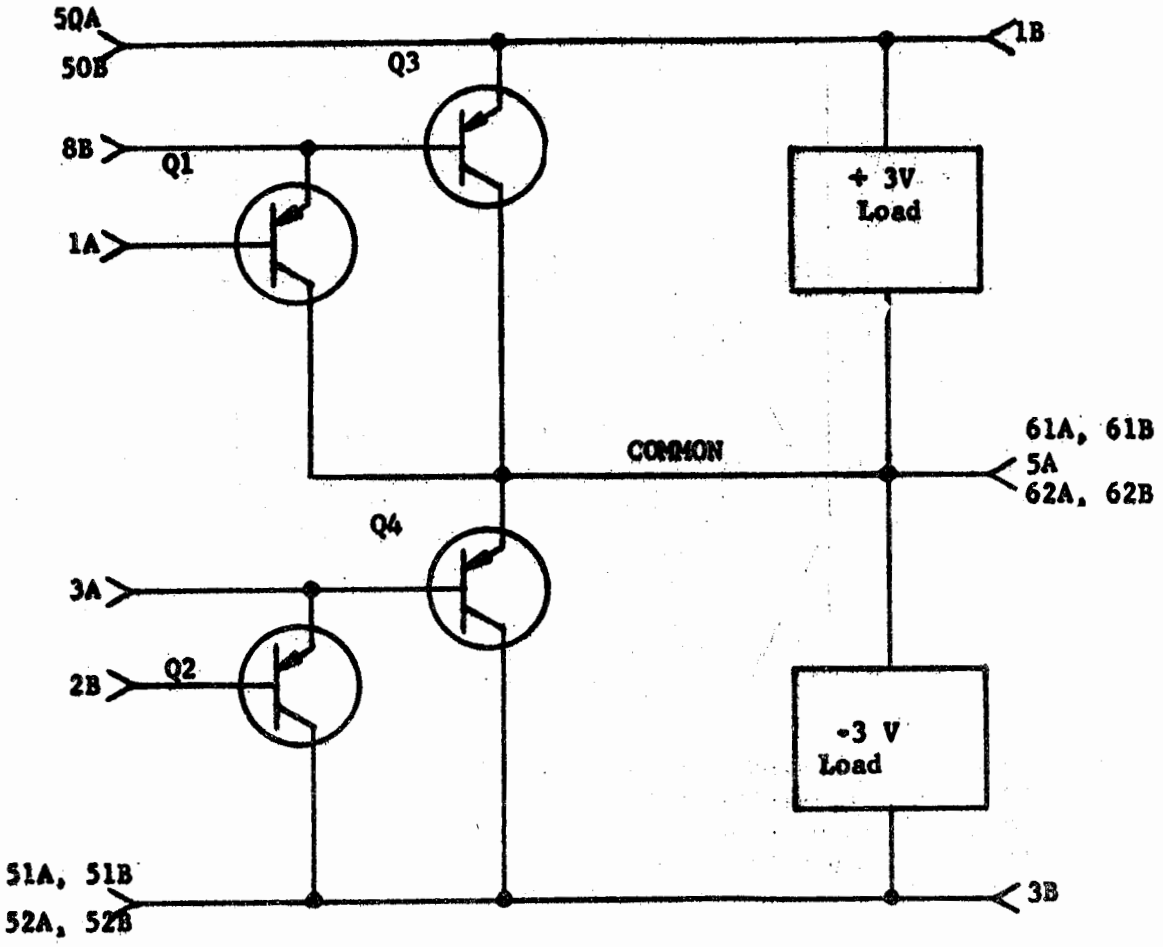


Q1: 10525764 Transistor - Medium Power
Q2: 10525749 Transistor - Power
Q3: 10525749 Transistor - Power
C1: 10525817 Capacitor - Tantalytic
C2: 10525816 Capacitor - Tantalytic

Terminal Numbers refer to those on the voltage regulator

FIGURE 2. *External circuitry: + 6 volt regulator.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:20Z
Check the source to verify that this is the current version before use.



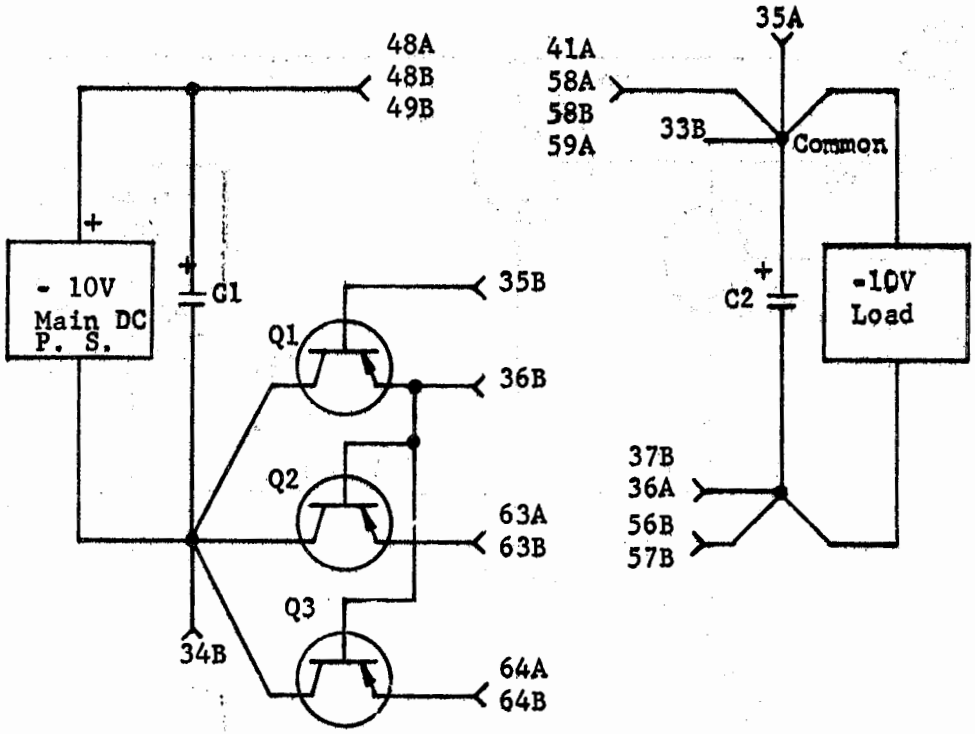
Q1, Q2: 10525764 Transistor - Medium Power

Q3, Q4: 10525749 Transistor - Power

Terminal numbers refer to those on the voltage regulator

FIGURE 3. *External circuitry: ± 3v regulator.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:20Z
Check the source to verify that this is the current version before use.

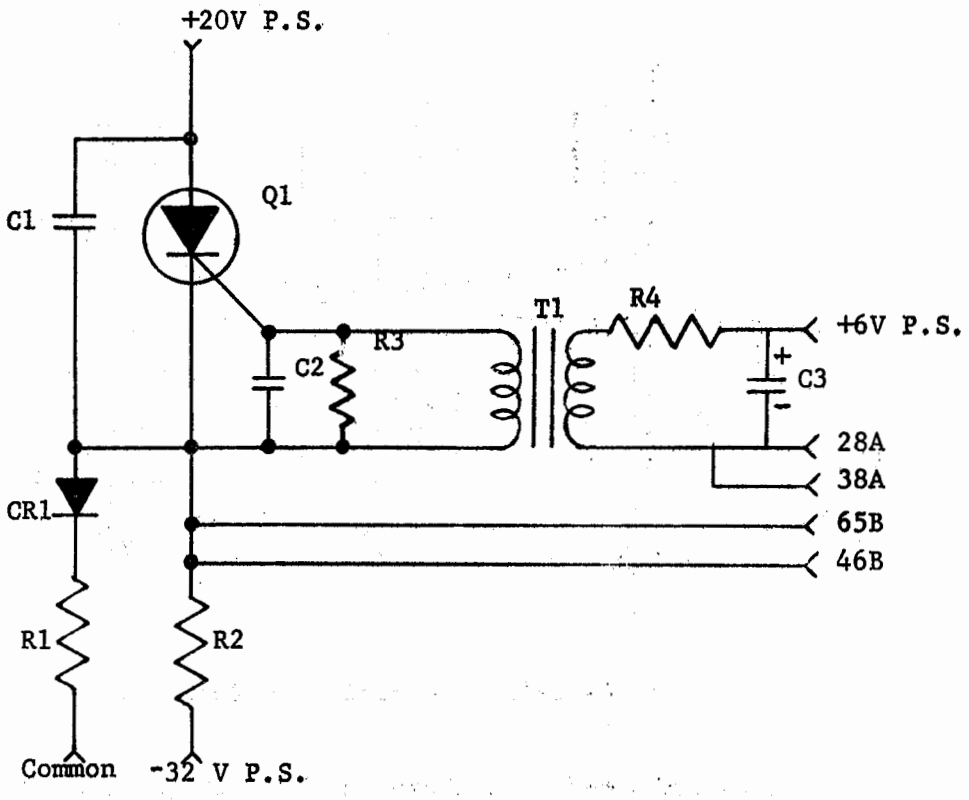


Q1: 10525764 Transistor - Medium Power
Q2: 10525749 Transistor - Power
Q3: 10525749 Transistor - Power
C1: 10525817 Capacitor - Tantalytic
C2: 10525816 Capacitor - Tantalytic

Terminal numbers refer to those on the voltage regulator

FIGURE 4. *External circuitry: — 10 v regulator.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:20Z
Check the source to verify that this is the current version before use.

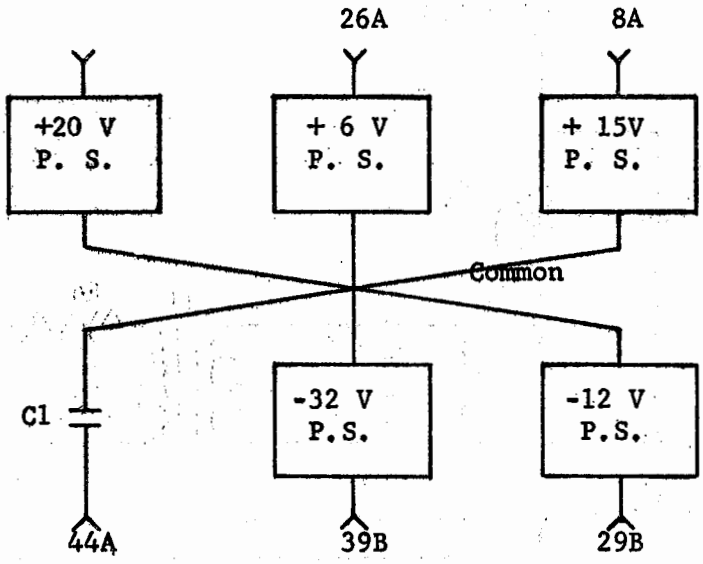


Q1: 10525779 Rectifier, Silicon Controlled
C1: 10525829 Capacitor - Mylar
C2: 10525810 Capacitor - Fusion, Sealed
C3: 10525824 Capacitor - Fixed Solid Electrolyte
R1: 10525778-5 Resistor - Power, Wire Wound
R2: 10526649 Resistor
R3: MS35043-87 Resistor, Fixed, Composition
R4: 10526600 Resistor - Fixed, Wire wound
T1: 10525744 Transformer - Pulse
CR1: 10525737 Diode - Silicon, Medium Power

Terminals refer to those on the voltage regulator

FIGURE 5. *External circuitry: master trip circuit.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:20Z
Check the source to verify that this is the current version before use.



C1: 10525817 - Capacitor - Tantalytic

Terminals refer to those on the voltage regulator

FIGURE 6. *External circuitry: power supplies.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:20Z
Check the source to verify that this is the current version before use.